Assessment of Thermoplastic Polyurethane (TPU) as a Chassis Material for 6-lb Combat Robotics

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Abstract

Combat robotics has a unique requirement for structural materials due to repeated high-energy impacts, the need for stiffness, and precise alignment of drive and weapon systems. Thermoplastic polyurethane (TPU) is widely used in robotics for bumpers, flexible armor, and energy-dissipating parts. However, the idea of building the primary chassis of a 6-pound combat robot entirely from TPU has significant drawbacks and concerns. This report evaluates the mechanical properties of TPU relative to conventional alternatives, potential failure modes, and provides recommendations for potential rigid material to TPU composites.

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1 Introduction

Combat robotics competitions, especially in the 6-pound weight class, require trade-offs and balance between structural strength, impact resistance, and weight efficiency. The chassis functions as a base on which all other parts are built, such as holding motors, batteries, controllers, and weapon systems in precise alignment. Conventionally, chassis are made of materials such as aluminum, polycarbonate, and carbon fiber composites due to their stiffness and predictable failure characteristics. Although TPU offers certain advantages, this paper serves to argue that it is not suitable as the only chassis material for competitive combat robots.

2 Material Properties of TPU

TPU is a flexible thermoplastic elastomer known for its high elasticity, abrasion resistance, and energy absorption capacity. Typical material hardness rating values range from 85A to 95A, corresponding to a soft and compliant structure compared to engineering plastics or metals. Its tensile strength (20–50 MPa) and Young's modulus (10–100 MPa) are orders of magnitude lower than aluminum (tensile strength $\tilde{3}$00 MPa, modulus 70 GPa). These differences immediately highlight the differences in stiffness and rigidity of TPU as a chassis material.[2]

2.1 Stiffness and Deflection

The primary requirement of a chassis is rigidity. The misalignment of the shafts, gears and wheels by even a few millimeters can compromise the performance of the drivetrain. The flexibility of TPU leads to significant elastic deformation under load. Consequently, weapon mounts and drivetrain geometry would be unstable in a chassis with TPU only.

2.2 Creep and Permanent Deformation

TPU exhibits creep behavior, meaning it undergoes time-dependent strain under constant stress. In a combat robot, where motors and weapons exert continuous torque and clamping loads, TPU would experience creep, leading to permanent warping or misalignment over time.

3 Structural Limitations in Combat Environments

3.1 Impact Response

While TPU absorbs impact energy effectively, this comes at the cost of structural integrity. Rigid chassis materials distribute impact loads to the entire frame, whereas TPU would localize deformations, potentially causing electronics and batteries to shift. Unlike sacrificial bumpers, which can be replaced after a match, the chassis are significantly difficult to change out if a part were to break. [1]

3.2 Fastener Retention

Combat robots rely on bolts, nuts, and inserts to secure components. TPU lacks the hardness necessary to retain threads under repeated shock loading. Heat-set inserts or embedded nuts often pull out during high-stress events. This poses significant concerns to serviceability and repair, key factors in competitions where rapid turnaround is required.

3.3 Thermal Concerns

TPU softens near 80–100°C. Motors, ESCs, and weapon systems can produce localized heating that approaches or exceeds this range. A TPU chassis could deform around hot components, further compromising precision and durability. [2]

4 Comparison with Conventional Materials

4.1 Aluminum

Aluminum alloys such as 6061-T6 provide an excellent balance of stiffness, machinability, and toughness. At modest thicknesses (2–4 mm), they offer rigidity far exceeding TPU at equivalent weight.

4.2 Polycarbonate

Polycarbonate provides impact resistance and resilience with greater stiffness than TPU. It can deform plastically without shattering, making it a common choice for side panels and structural plates in lightweight classes.

4.3 Carbon Fiber and G10

Composite materials provide high stiffness-to-weight ratios. While brittle compared to metals, they resist flexing and maintain precise geometry. These are often used in competitive designs where weight savings are critical.

5 Hybrid Approaches

Although TPU is unsuitable for full chassis use, it plays an important role as an impact absorption material. Hybrid approaches could include:

- TPU bumpers mounted to an aluminum or polycarbonate frame to absorb impact energy.
- TPU electronic mounts to decouple electrical system motion from motor vibration.
- TPU overlays that protect against abrasion while maintaining underlying structural stiffness.

These methods combine TPU's strengths with the rigidity of conventional materials.

6 Case Studies and Observations

Competitors in the 3-lb and 6-lb classes report frequent TPU failures when overused in structural applications. TPU-only bots often suffer drivetrain misalignment, delamination at fastener interfaces, and catastrophic deformation after encounters with spinners. In contrast, hybrid builds demonstrate longer service life and easier post-match maintenance.

- **Drivetrain misalignment / motors rubbing into chassis.** In a post-match analysis, Bone Dead Robotics noted that "The TPU chassis was too compliant... the outrunner motors outer body kept rubbing into parts of the chassis." This resulted in robot failure not from impacts, but from the motor tearing its own wires out and breaking down internally. [3]

7 Conclusion

The evaluation shows that while TPU is important as an impact-absorbing accessory, it is fundamentally unsuitable as the sole material for a 6-lb combat robot chassis. Its low stiffness, creep behavior, fastening challenges, and susceptibility to thermal deformation outweigh its advantages. The optimal strategy is a hybrid design employing rigid materials for structure and TPU for damping and protection.

A Notable TPU Robot Mention

The effectiveness of TPU-only chassis no longer becomes apparent for robots with weights above 3-lbs. For example, in the following video "**https://www.youtube.com/watch?v=x6V263T2r64**" shows a 12-lb combat robot using a lot of TPU for its components suffering from structural issues.

Some robots see success with TPU only chassis such as *Explosion* by Tommy Wong which has been noted as being extraordinarily tough and the most destructive in the beetleweights class (3 lbs). [3]

References

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[3] “Cattywampus-2024.” Bone Dead Robotics, 30 Dec. 2024, teambonedead.com/cattywampus-2024. Accessed 23 Sept. 2025.